CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

Traffic Assignment

Lecture Week 13

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Outline

- Introduction to traffic assignment / equilibrium
- Path choice models
  - Deterministic vs. Probabilistic
- Static traffic assignment
  - User Equilibrium (UE)
  - System Optimum (SO)
  - Methods to reach equilibrium
- Dynamic traffic assignment
  - Dynamic User Equilibrium (DUE),
  - Dynamic System Optimum (DSO)
- ~~Departure time choice~~

FINAL EXAM

Final Exam

- **All materials**
- **Exam time - *Internal***
  - – Examination Duration: **120** minutes
  - – Reading Time: **10** minutes

Final Exam

- **20** MCQ **– 50pts**
- **4** Short calculation questions – **50pts**
  - **Short calculation questions**
- Closed-book
- Permitted materials
  - One A4 sheet of handwritten or typed notes double sided is permitted
  - Unannotated Bilingual dictionary

Final Exam

- **MCQ**
  - **"Which one of the following statements is true/false?" - 15 questions**
  - **Brief calculation questions – 5 questions**
  - **Understand the concept, do not memorise!**
  - **2.5pts / question**
  - **5 choices for each question**

Final Exam

- Short calculation questions
  - Expect a balanced distribution across the modules
    - Car following,
    - Shockwaves,
    - Traffic assignment,
    - etc.
  - 4 questions in this format
  - Each worth 10-15 pts

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Course Teaching Feedback

- What are the benefits of completing the survey?
  - This is your opportunity to provide feedback on your learning experience! Your responses can be used to improve course design and quality. Please provide specific examples where possible.
  - Respectful constructive feedback is always helpful to help improve your learning experience. All responses submitted are strictly confidential.
- How do I complete the survey?
- Link

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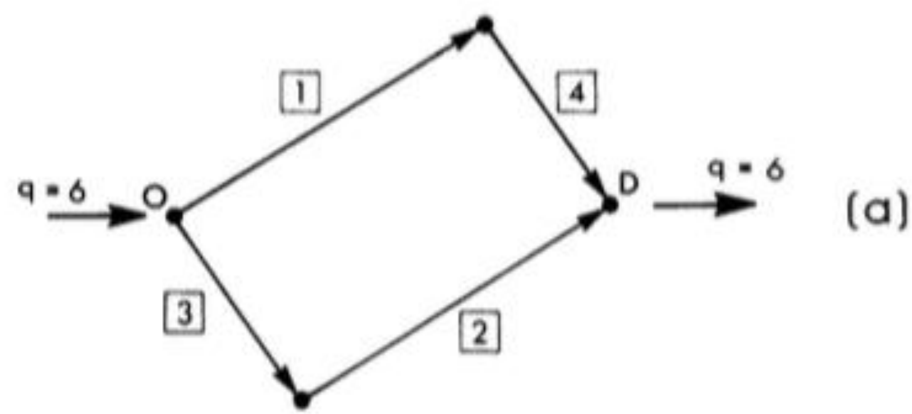
Static traffic assignment

Example of UE and SO calculation



Performance Data

$t_1(x_1) = 50 + x_1$

$t_2(x_2) = 50 + x_2$

$t_3(x_3) = 10x_3$

$t_4(x_4) = 10x_4$

Path Definition

Path 1 

Path 2 

Example of UE calculation

- The answers to the problems are
  - The four travel times t_1, t_2, t_3, t_4
  - The four link volumes x_1, x_2, x_3, x_4

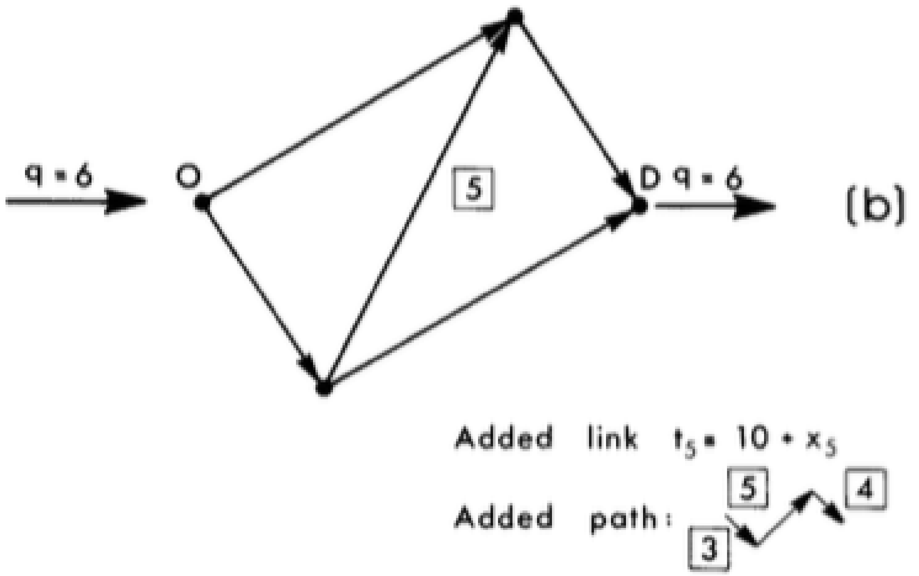
- User equilibrium means that the travel time on the two routes is equal
  - In the specific case $t_1 + t_4 = t_2 + t_3$
  - Write the travel times as a function of the four volumes x_1, x_2, x_3, x_4 given the link performance functions
  - Write the conservation of the traffic volumes at the nodes (in the specific case $x_1 = x_4$ and $x_2 = x_3$)
  - Write the total demand (in the specific case $x_1 + x_2 = 6$)
  - This is a system of four equations in four unknowns that can be easily solved

Example of SO calculation

- The answers to the problems are
  - The four travel times t_1, t_2, t_3, t_4
  - The four link volumes x_1, x_2, x_3, x_4

- System optimum means that the total travel time on the network needs to be written as a function of time and volumes and then be minimised
  - In the specific case $t_1x_1 + t_2x_2 + t_3x_3 + t_4x_4$
  - Write the travel times as a function of the four volumes x_1, x_2, x_3, x_4 given the link performance functions
  - Write the conservation of the traffic volumes at the nodes (in the specific case $x_1 = x_4$ and $x_2 = x_3$)
  - Write the total demand (in the specific case $x_1 + x_2 = 6$)
  - This is a function with three equations and four unknowns
  - The minimum is obtained by deriving the resulting function by the only unknown left and having the derivative equal to zero

Example of UE and SO calculation



Example of UE and SO calculation

- In the second case the total travel time in the network under UE conditions was worse even though the demand was the same and the network improved: this is the **Braess paradox** and it is related to the travellers being selfish

- From a more general perspective, **Braess paradox** emphasizes the importance of a careful and systematic analysis of investments in urban networks. Not every addition of capacity can bring about all the anticipated benefits and in some cases, the situation may be worsened.

Solution methods

Method of successive averages (MSA)

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- **Step 0:** *Initialization*. Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on t_a = $t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter n = 1
- **Step 1:** *Update.* $t_a^n = t_a(x_a^n)$
- **Step 2:** *Direction finding.* Perform all-or-nothing assignment based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3:** *Define an updating parameter.* Usually $\phi = 1/n$.
- **Step 4:** *Move.* $x_a^{n+1} = x_a^n + \phi(y_a^n - x_a^n)$, $\forall a$
- **Step 5:** *Convergence check.* If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

$$\|x_a^{n+1} - x_a^n\| \leq \varepsilon$$

Frank-Wolfe Assignment

- **Step 0:** *Initialization*. Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on t_a = $t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter n = 1
- **Step 1:** *Update.* $t_a^n = t_a(x_a^n)$
- **Step 2:** *Direction finding.* Perform all-or-nothing assignment based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3:** *Line search.* Find α_n that solves:

$$\min_{0\leq\alpha\leq 1} \sum_a \int_0^{x_a^n+\alpha(y_a^n-x_a^n)} t_a(w)dw$$

- **Step 4:** *Move.* $x_a^{n+1} = x_a^n + \alpha_n(y_a^n - x_a^n)$, $\forall a$
- **Step 5:** *Convergence check.* If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

MSA – Stochastic User Equilibrium

- **Step 0:** *Initialization*. Select initial link costs: usually free flow travel times. Perform all-or-nothing assignment based on t_a = $t_a(0)$, $\forall a$. This yields $\{x_a^1\}$. Set counter n = 1
- **Step 1:** *Update.* $t_a^n = t_a(x_a^n)$
- **Step 2:** *Direction finding.* Perform a stochastic assignment based on $\{t_a^n\}$. This yields a set of auxiliary flows $\{y_a^n\}$.
- **Step 3:** *Define an updating parameter.* Usually $\phi = 1/n$.
- **Step 4:** *Move.* $x_a^{n+1} = x_a^n + \phi(y_a^n - x_a^n)$, $\forall a$
- **Step 5:** *Convergence check.* If a convergence criterion is met, stop (the current solution, $\{x_a^{n+1}\}$, is the set of equilibrium link flows); otherwise, set $n = n + 1$ and go to step 1.

$$\|x_a^{n+1} - x_a^n\| \leq \varepsilon$$

Dynamic traffic assignment (DTA)

The problems with static assignment

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For example...

1. Not all vehicles will experience the same delay
2. What does the flow mean in static assignment?
3. Where does congestion actually lie?
4. What about queue spillback?

$$t = t_f \left(1 + \alpha \left(\frac{v}{c}\right)^{\beta}\right)$$

What this means is that we need a fundamentally different congestion model, based in traffic flow theory rather than arbitrary functions.

Dynamic extension

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Dynamic User Equilibrium (DUE)

- All routes used by travellers leaving the same origin **at the same time** for the same destination have equal and minimal travel time.

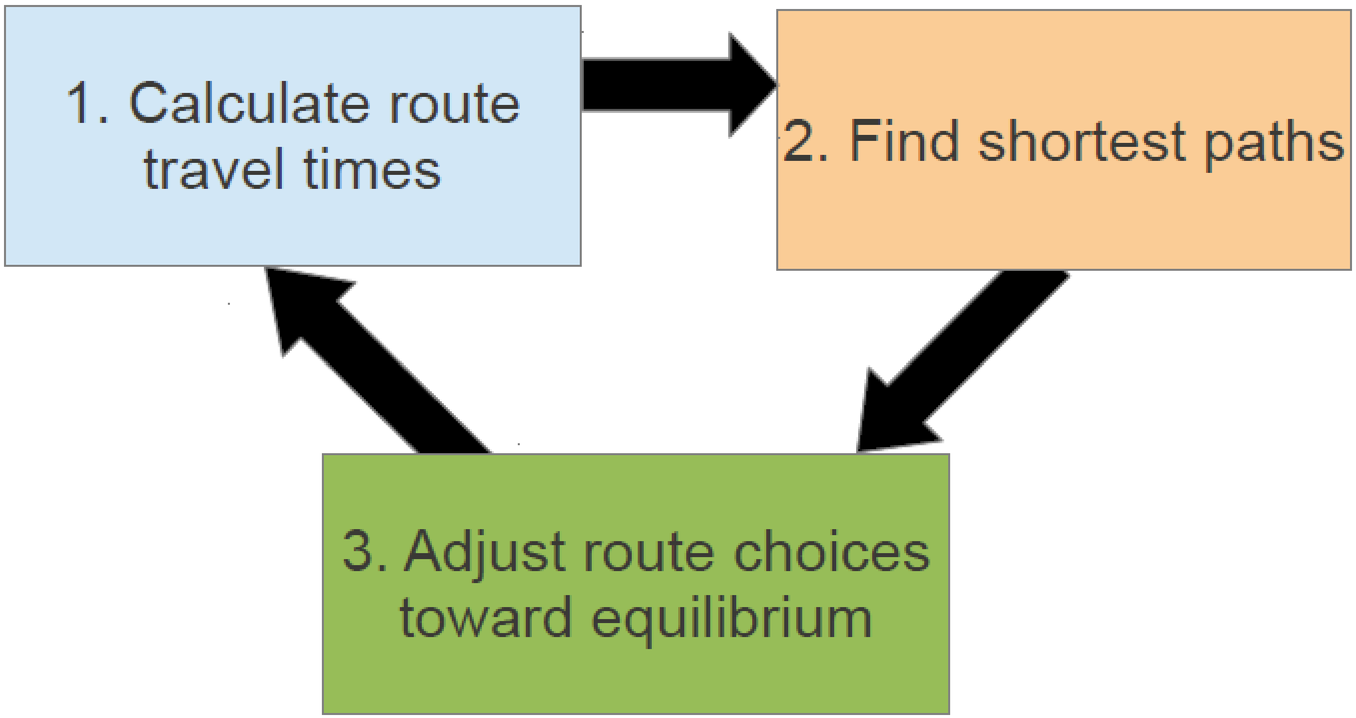
Essential requirements

- A model for how congestion (travel times) varies over time.
- A concept of equilibrium route choice.
- Equilibration based on experienced travel times, not instantaneous travel times.

What most simulation packages including Aimsun does is a one-shot assignment, not DTA!! Although it is presented as DTA..

DTA

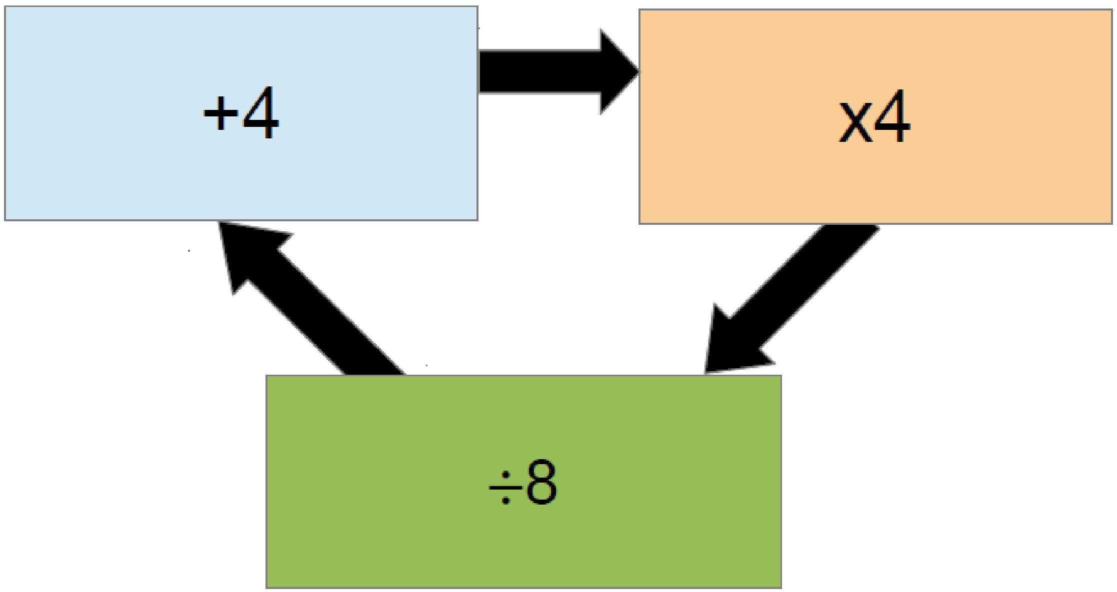
A similar iterative procedure to static traffic assignment



Interestingly, the easiest step in static (updating path travel times) is the hardest one in dynamic.

DTA – Fixed point

- Fixed-point solution can be presented in a mathematical form

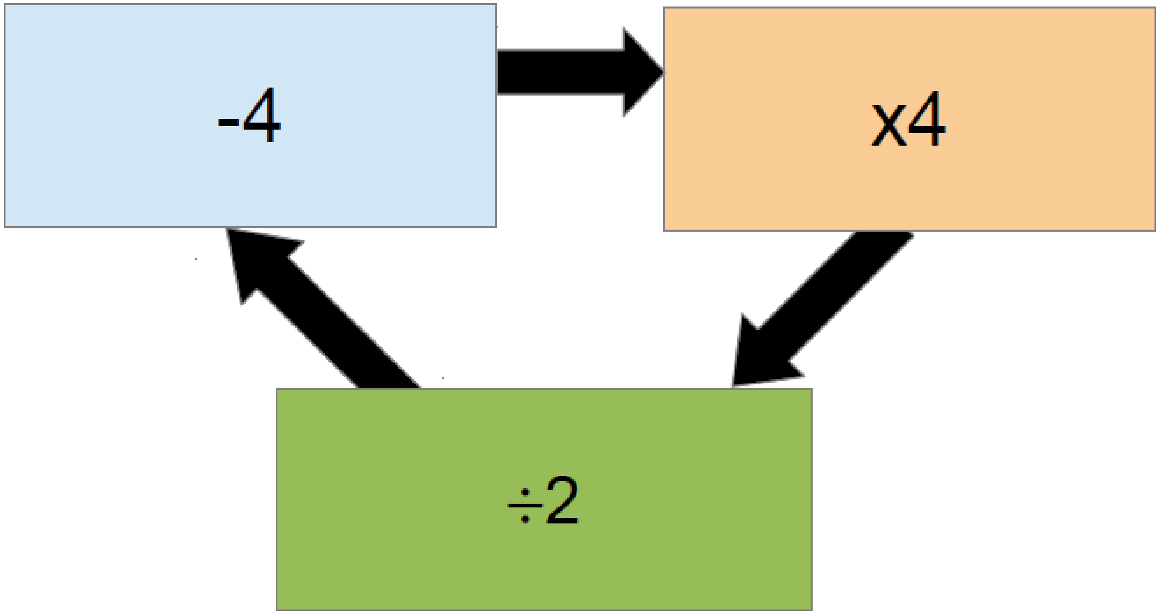


- This problem can be solved by denoting the values in the boxes as x, y, and z.
- We then have $y = x + 4$, $z = 4y$, and $x = z/8$. Substituting them into each other, we have $x = 0.5x + 2$ or $x = 4$.

DTA – Fixed point

- A not-so-clever approach is to pick a starting value, then calculate through the loop iteratively.
- 10 → 7 → 5.5 → 4.75 → 4.375 → … which converges to the correct answer (4).
- Picking a different starting value: 1 → 2.5 → 3.25 → 3.625 → … also converges to the same answer.
- No convergence guarantee particularly in DTA!!

DTA – Fixed point



- Choosing the same starting value, we have 10 → 12 → 16 → 24 → 40 →… which diverges to +Inf.
- The clever approach still works: y = x - 4, z = 4y, and x = z/2, so x = 2x - 8 and x = 8. If we used this as our starting value, the simple approach would work, but for any other value it will diverge.

DTA – Fixed point

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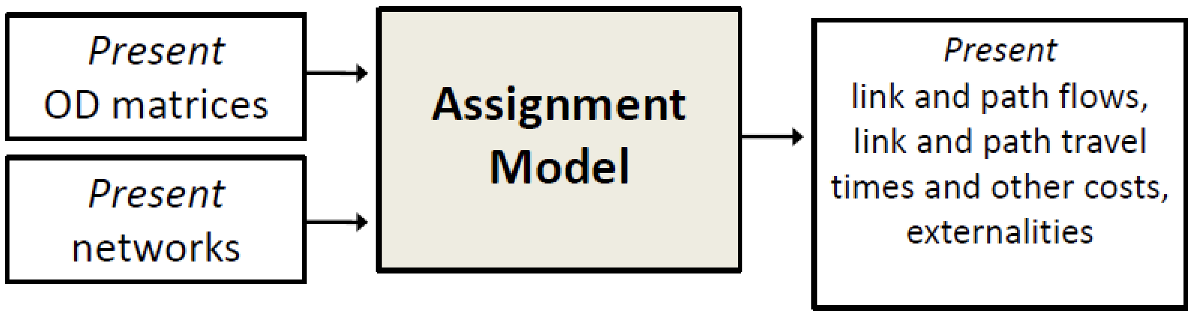
- **The moral of the story:** if we are simply iterating between the three steps, we need to be careful to ensure that the solution will converge to a consistent answer.
- DTA problems in large-scale networks are complicated enough that **we do not know of a faster or a clever method**.
- We are stuck with iterative methods, which suffer significant convergence issues particularly in congested large-scale networks.

Applications

Applications of Assignment models

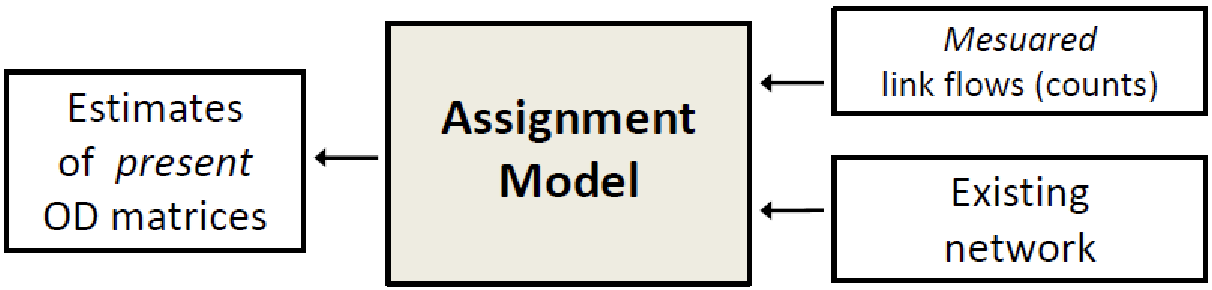
- As estimators of the current state of the transport systems



- A good match between observed and simulated traffic conditions is needed to guarantee a well calibrated simulation model
- The results from the assignment model can complement other traffic observations (e.g., link flows, path travel time measurements, etc.), as such measures are not available for all network components

Applications of Assignment models

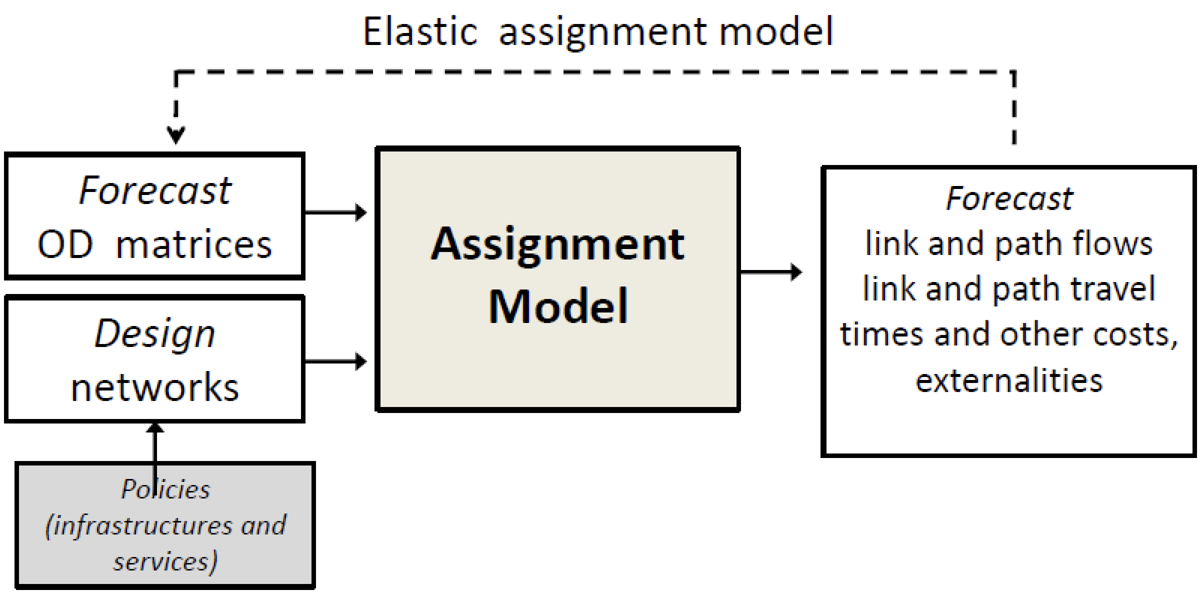
- For the estimation of current travel demand



- If prior OD matrices are not reliable, assignment models can be useful to estimate present OD matrices
- Prior ODs usually depend on historical surveys and may not reflect current traffic state

Applications of Assignment models

- For evaluating the changes in the transport systems



Next

This week

- Tutorial